THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE UNIT TEST SET A

Chapters 9–12: **Light – Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. A lens produces a virtual, erect and magnified image. The lens is: 1

(a) concave **(b)** plane glass
(c) convex, object between F and O **(d)** convex, object beyond 2F

2. A ray passing through the principal focus of a concave mirror after reflection: 1

(a) passes through F again **(b)** retraces its path
(c) becomes parallel to the principal axis **(d)** passes through C

3. A student states: “The refractive index of diamond (2.42) means light travels fastest in it.” The correct response is: 1

(a) Correct
(b) Correct only for red light
(c) Incorrect — it is optically the densest; light is slowest
(d) Cannot be determined

4. The opening through which light enters the eye is the: 1

(a) iris **(b)** lens **(c)** pupil **(d)** retina

5. The image formed on the retina of the eye is: 1

(a) virtual and inverted **(b)** real and inverted
(c) virtual and erect **(d)** real and erect

6. Myopia can be corrected by using a: 1

(a) cylindrical lens **(b)** concave lens **(c)** convex lens **(d)** bifocal lens

7. The number of electrons in 1 coulomb of charge is about: 1

(a) 6.25×10^{18} **(b)** 6.25×10^{19} **(c)** 1.6×10^{19} **(d)** 1.6×10^{-19}

8. The SI unit of resistivity is: 1

(a) ohm **(b)** ohm metre **(c)** volt **(d)** ohm per metre

9. A charge of 10 C flows through a wire in 5 s. The current is: 1

(a) 15 A **(b)** 0.5 A **(c)** 50 A **(d)** 2 A

10. For the direction of the field around a straight wire, Student A uses the right-hand thumb rule; Student B uses Fleming's left-hand rule. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

11. The force on a current-carrying conductor in a magnetic field is maximum when the conductor is: 1

(a) perpendicular to the field **(b)** parallel to the field
(c) at 45° **(d)** outside the field

12. A student states: “Two magnetic field lines never intersect.” The correct response is: 1

(a) Correct **(b)** Only near poles

(c) Incorrect **(d)** Cannot be determined

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. A student says that a concave lens is used in a magnifying glass. Justify or refute. 2

14. What is presbyopia? 2

15. What is the function of the iris? 2

16. Why is tungsten used for filaments of bulbs? 2

17. State the right-hand thumb rule. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. State Snell's law of refraction. The refractive index of glass with respect to air is 1.5. What is the refractive index of air with respect to glass? 3

19. Draw (describe) a ray diagram to show how a concave lens corrects myopia. 3

20. A wire of resistance 10 Ω is stretched to double its length. Find its new resistance (volume constant). 3

21. What precautions should be taken to avoid overloading and short-circuiting? 3

22. A current-carrying conductor is placed in a magnetic field. When is the force on it 3

(i) maximum

(ii) zero? Explain the force direction using a rule.

OR

Why is earthing of appliances necessary? Which colour wire is used? 3

23. Write three uses of concave mirrors. 3

OR

The refractive indices of kerosene, turpentine and water are 1.44, 1.47 and 1.33. In which does light travel fastest? Which is optically densest? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

24. (i) What is hypermetropia? 5

(ii) Describe with ray diagrams the defect and its correction.

(iii) What are the causes?

OR

(i) What is dispersion? 5

(ii) Why do different colours deviate differently?

(iii) Describe Newton's experiment with two prisms.

(iv) What is a spectrum?

25. (i) Describe an activity to show that a current-carrying conductor experiences a force in a magnetic field. 5

(ii) How does the direction of force change if the current is reversed?

(iii) Name the rule used to find the direction.

OR

(i) State Fleming's left-hand rule with a diagram. 5

(ii) A proton moves in a magnetic field directed into the page; it moves to the right. Find the direction of force.

(iii) Name two devices using this effect.

END OF QUESTION PAPER Total: 50 Marks